

Computational Companion to “Generalized Quasi-Symmetric Nets: A Constructive Approach to the Equimodular Elliptic Type of $n \times m$ Kokotsakis Polyhedra” — Helper

A. Nurmatov, M. Skopenkov, F. Rist, J. Klein, D. L. Michels
Tested on: Mathematica 14.0

Section 1

Results of Section 1 are used in the proof of Lemma 8 (see paper)

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In[55]:= (* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)
ClearAll[ $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\sigma$ ,  $\epsilon$ , a, b, c, d, r, s, f, M, rad1,
  sq1, assumptions1, isTrueQ1, isTrueQeps1, row1, identities1];
 $\sigma = (\alpha + \beta + \gamma + \delta) / 2$ ;
a = Sin[ $\alpha$ ] / Sin[ $\sigma - \alpha$ ]; b = Sin[ $\beta$ ] / Sin[ $\sigma - \beta$ ];
c = Sin[ $\gamma$ ] / Sin[ $\sigma - \gamma$ ]; d = Sin[ $\delta$ ] / Sin[ $\sigma - \delta$ ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain
  assumptions:the flat angles and the sigma-shifted angles lie in (0,Pi);
  these hold under the hypotheses of the lemma
  indicated in the subsection above (see paper)*)
assumptions1 = {0 <  $\alpha$  < Pi, 0 <  $\beta$  < Pi, 0 <  $\gamma$  < Pi,
  0 <  $\delta$  < Pi, 0 <  $\sigma - \alpha$  < Pi, 0 <  $\sigma - \beta$  < Pi, 0 <  $\sigma - \gamma$  < Pi, 0 <  $\sigma - \delta$  < Pi};
rad1 = (f - M) (r - 1) (s - 1) / (f (1 - M));
sq1 = (Sin[ $\sigma$ ] / Sin[ $\sigma - \delta$ ]) ^2;
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isTrueQ1[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions1]];
(*Two-stage verification,
in the order of the table:STAGE 1 (the radicand row) identifies rad1 with the
perfect square sq1.STAGE 2 (the identity row) uses Stage 1:for t=
Sin[sigma]/Sin[sigma-delta],the eps-split assumptions
give t the sign eps (u/v checks),
so Sqrt[rad1]=Sqrt[t^2]=eps*t (abstract root law),
and eps*Sqrt[rad1]*Sin[sigma-delta]=
eps^2*Sin[sigma]=Sin[sigma] (cancellation).No Sqrt
over trigonometric arguments reaches the kernel.*)
isTrueQeps1[lhs_, rhs_] :=
  Module[{t, u, v}, TrueQ[FullSimplify[rad1 == sq1, Assumptions → assumptions1]] &&
    TrueQ[Assuming[u > 0 && v > 0, Simplify[u / v > 0]]] &&
    TrueQ[Assuming[u < 0 && v > 0, Simplify[u / v < 0]]] &&
    TrueQ[Assuming[t > 0, Simplify[Sqrt[t^2] == t]]] &&
    TrueQ[Assuming[t < 0, Simplify[Sqrt[t^2] == -t]]] &&
    TrueQ[Simplify[(Sin[sigma] / Sin[sigma - delta]) Sin[sigma - delta] == Sin[sigma]]];
(* =====*)
(*PART 2:THE RADICAND AND THE IDENTITY*)
(* =====*)
identities1 = {"Radicand equals a perfect square",
  HoldForm[(f - M) (r - 1) (s - 1) / (f (1 - M))],
  HoldForm[(Sin[sigma] / Sin[sigma - delta])^2], isTrueQ1,
  {"Identity for sin(sigma) (via the radicand row)", HoldForm[Sin[sigma]],
  HoldForm[epsilon Sqrt[(f - M) (r - 1) (s - 1) / (f (1 - M))] Sin[sigma - delta]], isTrueQeps1}};
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row1[{desc_, lhs_, rhs_, test_}] :=
  {desc, Item[TraditionalForm[lhs], Alignment → Center],
  Item[TraditionalForm[rhs], Alignment → Center],
  If[test[ReleaseHold@lhs, ReleaseHold@rhs],
  Style["Verified", Darker@Green, Bold], Style["FAILED", Red, Bold]]};
Grid[Join[{{Style["Description", Bold], Style["Left Hand Side (LHS)", Bold],
  Style["Right Hand Side (RHS)", Bold], Style["Equality Status", Bold]}},
  row1 /@ identities1], Frame → All, Background → {None, {1 → LightGray}},
  Spacings → {2, 2.5}, Alignment → {{Left, Center, Center, Center}},
  ItemStyle → Directive[FontFamily → "Helvetica", 12]]

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Out[67]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Radicand equals a perfect square	$\frac{(f-M)(r-1)(s-1)}{f(1-M)}$	$\left(\frac{\sin(\sigma)}{\sin(\sigma-\delta)}\right)^2$	Verified
Identity for $\sin(\sigma)$ (via the radicand row)	$\sin(\sigma)$	$\varepsilon \sqrt{\frac{(f-M)(r-1)(s-1)}{f(1-M)}} \sin(\sigma - \delta)$	Verified

Section 2

Results of Section 2 are used in the proof of Lemma 12 (see paper)

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In[68]:= (* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)
ClearAll[α, β, γ, δ, σ, ε, a, b, c, d, r, s, f, M, assumptions2,
  isTrueQ2, isTrueQeps2, row2, radicands2, identities2];
σ = (α + β + γ + δ) / 2;
a = Sin[α] / Sin[σ - α]; b = Sin[β] / Sin[σ - β];
c = Sin[γ] / Sin[σ - γ]; d = Sin[δ] / Sin[σ - δ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain assumptions:the flat angles lie in (0,Pi),
as in the hypotheses of the lemma indicated in the subsection above*)
assumptions2 = {0 < α < Pi, 0 < β < Pi, 0 < γ < Pi, 0 < δ < Pi};
(*Testing functions;
the fourth identity is tested over both signs of Sin[sigma],
that is,over eps=+-1*)
isTrueQ2[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions2]];
isTrueQeps2[lhs_, rhs_] := TrueQ[FullSimplify[lhs == (rhs /. ε → +1),
  Assumptions → Append[assumptions2, Sin[σ] > 0]]] && TrueQ[FullSimplify[
  lhs == (rhs /. ε → -1), Assumptions → Append[assumptions2, Sin[σ] < 0]]];
(* =====*)
(*PART 2:THE RADICANDS AND THE FOUR IDENTITIES*)
(* =====*)
radicands2 = {"Radicand 1 equals a perfect square",
  HoldForm[f (f - M) (s - 1) / (s (s - M) (f - 1))], HoldForm[(Sin[α] / Sin[δ]) ^ 2],
  isTrueQ2}, {"Radicand 2 equals a perfect square",
  HoldForm[M ^ 2 (r - 1) (s - 1) / (r s (r - M) (s - M))],
  HoldForm[(Sin[β] / Sin[δ]) ^ 2], isTrueQ2},

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{"Radicand 3 equals a perfect square",
 HoldForm[f (f - M) (r - 1) / (r (r - M) (f - 1))], HoldForm[(Sin[γ] / Sin[δ]) ^2],
 isTrueQ2}, {"Radicand 4 equals a perfect square",
 HoldForm[(f - M) (r - 1) (s - 1) / (r s (1 - M))],
 HoldForm[(Sin[σ] / Sin[δ]) ^2], isTrueQ2}};
identities2 = {"Identity 1", HoldForm[Sin[α]],
 HoldForm[Sqrt[f (f - M) (s - 1) / (s (s - M) (f - 1))] Sin[δ]],
 isTrueQ2}, {"Identity 2", HoldForm[Sin[β]],
 HoldForm[Sqrt[M^2 (r - 1) (s - 1) / (r s (r - M) (s - M))] Sin[δ]],
 isTrueQ2}, {"Identity 3", HoldForm[Sin[γ]],
 HoldForm[Sqrt[f (f - M) (r - 1) / (r (r - M) (f - 1))] Sin[δ]],
 isTrueQ2}, {"Identity 4", HoldForm[Sin[σ]],
 HoldForm[ε Sqrt[(f - M) (r - 1) (s - 1) / (r s (1 - M))] Sin[δ]], isTrueQeps2}};
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row2[{desc_, lhs_, rhs_, test_}] :=
 {desc, Item[TraditionalForm[lhs], Alignment → Center],
 Item[TraditionalForm[rhs], Alignment → Center],
 If[test[ReleaseHold@lhs, ReleaseHold@rhs],
 Style["Verified", Darker@Green, Bold], Style["FAILED", Red, Bold]]};
Grid[Join[{{Style["Description", Bold], Style["Left Hand Side (LHS)", Bold],
 Style["Right Hand Side (RHS)", Bold], Style["Equality Status", Bold]}},
 row2 /@ Join[radicands2, identities2]], Frame → All,
 Background → {None, {1 → LightGray}}, Spacings → {2, 2.5},
 Alignment → {{Left, Center, Center, Center}},
 ItemStyle → Directive[FontFamily → "Helvetica", 12]]

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Out[79]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Radicand 1 equals a perfect square	$\frac{f(f-M)(s-1)}{s(s-M)(f-1)}$	$\left(\frac{\sin(\alpha)}{\sin(\delta)}\right)^2$	Verified
Radicand 2 equals a perfect square	$\frac{M^2(r-1)(s-1)}{r s(r-M)(s-M)}$	$\left(\frac{\sin(\beta)}{\sin(\delta)}\right)^2$	Verified
Radicand 3 equals a perfect square	$\frac{f(f-M)(r-1)}{r(r-M)(f-1)}$	$\left(\frac{\sin(\gamma)}{\sin(\delta)}\right)^2$	Verified
Radicand 4 equals a perfect square	$\frac{(f-M)(r-1)(s-1)}{r s(1-M)}$	$\left(\frac{\sin(\sigma)}{\sin(\delta)}\right)^2$	Verified
Identity 1	$\sin(\alpha)$	$\sqrt{\frac{f(f-M)(s-1)}{s(s-M)(f-1)}} \sin(\delta)$	Verified
Identity 2	$\sin(\beta)$	$\sqrt{\frac{M^2(r-1)(s-1)}{r s(r-M)(s-M)}} \sin(\delta)$	Verified
Identity 3	$\sin(\gamma)$	$\sqrt{\frac{f(f-M)(r-1)}{r(r-M)(f-1)}} \sin(\delta)$	Verified
Identity 4	$\sin(\sigma)$	$\varepsilon \sqrt{\frac{(f-M)(r-1)(s-1)}{r s(1-M)}} \sin(\delta)$	Verified